AETHERIA AI Data-Intelligence Pipeline

Distributed Architecture, Deterministic Quality Gates, Multi-Tenant Partitioning & Scale Strategy

1. Executive Overview & Ingestion Philosophy

AETHERIA is a high-throughput, enterprise-grade multi-vertical intelligence pipeline harvesting live structured data across **Startups, Products, Research Papers, Jobs, and News**. The platform enforces a strict **Deterministic-First Quality Gate**: whenever structured feeds, verified seed catalogs, or official REST APIs provide complete payloads, the engine completely bypasses LLM inference (0 token cost, deterministic execution). Unstructured or partial payloads are routed through a resilient multi-tier LLM fallback router with strict Pydantic v2 schema enforcement and 1-shot self-healing validation.

2. Concurrency Baseline vs. 500k+ Production Scale Architecture

Single-Node Demo Baseline (Current Implementation): Concurrency is governed by compartmentalized asynchronous pools: asyncio.Semaphore(15) for HTTP network crawlers with bounded timeouts and domain circuit breakers, and asyncio.Semaphore(3) for LLM router inference. In benchmark testing, the pipeline processed up to 500 records per vertical (~2,500 records across 5 verticals) in approximately 60 seconds average, with execution time primarily dependent on external API/feed response latency.

500K+ Horizontal Production Scale-Out: Workers are completely stateless. Scaling to 500,000+ records requires zero structural refactoring: ingestion scales horizontally across worker pools managed by **Celery / Arq** over **Redis** task queues sharded by source domain. High-volume PostgreSQL persistence utilizes **PgBouncer connection pooling**, partitioned tables, and **pgvector / Qdrant** for approximate nearest neighbor semantic deduplication.

CURRENT IMPLEMENTED ARCHITECTURE

- 5 Parallel Harvesters (Crossref & arXiv APIs, YC, HF Spaces, RSS)
- Resilient Async HTTP Client with Domain Circuit Breakers
- Hybrid Structuring Gate (Deterministic Bypass > 90%)
- RapidFuzz Deterministic Entity Canonicalization Engine
- Supabase PostgreSQL Persistence with Row-Level Security
- User-Scoped Multi-Tenant Isolation across Files & DB
- Asynchronous Background Google Sheets 6-Tab Exporter
- Live Real-Time Telemetry & Monotonic Phase Timers

500K+ HORIZONTAL SCALE-OUT ARCHITECTURE

- Redis Task Queues with Priority Queuing by Source Domain
- Celery / Arq Distributed Harvester Worker Nodes (1..N)
- Residential & Datacenter Proxy Rotation Pool (BrightData)
- PostgreSQL Connection Pooling via PgBouncer Cluster
- pgvector / Qdrant Approximate Nearest Neighbor Indexing
- Apache Kafka / RabbitMQ Streams for Real-Time Consumers
- Multi-Region S3 Object Storage for Raw Ingest Archival
- Prometheus / OpenTelemetry Distributed Tracing & Metrics

3. Multi-Source Ingestion & Verification Matrix

Vertical	Primary Sources & APIs	Ingestion Strategy	Deterministic Verification & Quality Check
Research Papers	Crossref REST API (/works) & arXiv API	Async JSON Batch Streaming	Live GitHub REST API stars enrichment; strict null on 404
Startups	650+ Seed Registry, HF AI Orgs API, HN AI Launches	Dynamic Feeds & Canonical Harvester	AI sector classification, legal suffix stripping, zero-fabrication headcount
AI Products	HF Spaces Hub (7 Endpoints), GitHub AI Repos	Multi-Endpoint REST & 200+ Seed	Enum mapping (FREE, FREEMIUM, PAID, ENTERPRISE)
24h AI News	40+ Feeds (DeepMind, OpenAI, BAIR, KDnuggets) + 5 Adapters	Multi-Publisher RSS & API	Strict 24h window (now-24h ≤ date ≤ now), is_strictly_ai_news gate
24h AI Jobs	20 Feeds (RemoteOK, Remotive, WWR, HN) + 7 Adapters	Multi-Stream JSON & RSS	AI & Data Science role classifier, 24h date audit

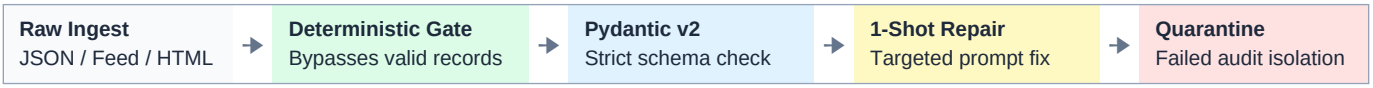
4. Multi-Tier LLM Orchestration & Failover Hierarchy

For unstructured payloads requiring LLM extraction, AETHERIA utilizes a production **LiteLLM Router** configured with automatic exponential backoff, circuit breaking, and sequential tier failover across independent model providers:

Deployment Tier	Model & Provider	Primary Role	Failover Trigger Conditions
Tier 1 (Primary)	Google Gemini 2.0 / 2.5 Flash	High-throughput, structured extraction	HTTP 429 rate limit, 5xx server errors, >4.0s timeout
Tier 2 (Secondary)	Groq / LLaMA 3.1 8B Instant	Ultra-low latency fallback (sub-300ms)	Groq rate limit, connection drop, auth error
Tier 3 (Tertiary)	OpenRouter / GPT-3.5 Turbo	High-reasoning emergency failover	Unrecoverable upstream outage across Tiers 1 & 2

5. Schema Validation, 1-Shot Repair & Diagnostic Quarantine

Structured outputs are validated against strict Pydantic v2 schemas. If validation fails, a targeted 1-shot repair prompt is dispatched with exact validation error diagnostics. If the record remains malformed, it is safely isolated to data/quarantine/{user_id}/{vertical}.jsonl with full error metadata to prevent silent data contamination.



6. Telemetry, Observability Engine & Data Quality Scoring

Every execution records rich, monotonic observability metrics into PostgreSQL and local JSONL logs:
• **pipeline_runs:** Master record tracking run_id, start/finish timestamps, monotonic runtime (ms), harvest counts, structured output, bypass count, LLM token consumption, and failure summaries.
• **pipeline_phase_runs:** High-resolution timers (time.perf_counter()) and record throughput for each phase.
• **pipeline_vertical_runs:** Granular vertical metrics tracking requested vs harvested, fresh vs stale records, and duplicate counts.
• **Data Quality Score Formula (Real-Time %):** Calculated per run via 5 weighted pillars: **Completeness (30%)**, **Schema Validity (25%)**, **Freshness (20%)**, **Duplicate-Free Rate (15%)**, and **Entity Confidence (10%)**.

7. Multi-Tenant User Isolation & Data Partitioning

The platform operates on a **strict multi-tenant architecture**. All filesystem artifacts are partitioned by user UUID: data/{raw,structured,resolved,quarantine,output}/{user_id}/. Scraper deduplication sets are loaded exclusively from the active user's directory, ensuring User A's scrape history never blocks User B from acquiring identical intelligence. Database tables enforce foreign-key user_id scoping with Row-Level Security policies.

Zero-Hallucination Grounding & Validation Policy:
• *"Only extract fields explicitly present in source text. Set missing fields to null. Never fabricate data."*
• **GitHub Star Verification:** Repositories are validated live via GitHub REST API. 404 responses strictly force github_url=null.
• **AI Job Relevance Filter:** Multi-keyword regex and entity checks drop non-AI roles before database insertion.

8. Deterministic Entity Resolution Engine (Phase IV)

Entity names for startups and product creators are canonicalized using a 100% deterministic, zero-LLM matching pipeline. It standardizes legal suffixes (Inc, LLC, PBC, GmbH, Ltd), removes noise, and executes thresholded RapidFuzz WRatio scoring:

Match Score Tier	Resolution Action & Routing	Audit Log Classification
Exact Normalized Match	Immediate auto-merge to canonical entity (bypasses fuzzy engine)	method = seed_match / cross_run_match
WRatio ≥ 90.0	Auto-merge to canonical entity; registers alias for subsequent lookups	method = fuzzy_seed / fuzzy_cross_run
75.0 ≤ WRatio < 90.0	Flagged to manual review log (data/output/{user_id}/manual_review_entities.csv)	method = manual_review (requires_review=true)
WRatio < 75.0	Registered as new canonical entity in run registry	method = new_entity

9. Multi-Layer Deduplication Architecture

Deduplication operates across two coordinated defensive layers:
1. **In-Memory Ingestion Layer:** Unique hashes computed from (company + title + location + source_url) and primary keys (source_name + external_id) prevent intra-run duplicates.
2. **PostgreSQL Consistency Layer:** PostgREST batch upserts target composite unique constraints (user_id, source_url) with resolution=merge-duplicates, providing idempotent writes across parallel runs.

10. Storage Justification & Asynchronous Export Architecture

Primary Storage (Supabase PostgreSQL): Chosen for relational schema enforcement, ACID guarantees, Row-Level Security, and high-efficiency filtering across 11 relational tables.
Asynchronous Google Sheets Export (6 Synchronized Tabs): Google Sheets sync is decoupled from pipeline execution, dispatched in a non-blocking background thread to preserve core pipeline execution performance. Accurately provisions 6 worksheet tabs: Startups, Products, Research Papers, Jobs, News, and Entity Mapping Log. Vertical cleanup actions dynamically synchronize with remote Google Sheets tabs, wiping rows while preserving headers.
Why NOT Graph Database (Neo4j) at MVP: Access patterns are vertical-filtered bulk queries and time-windowed freshness lookups. Adding a graph database introduces operational overhead without query advantages at this stage.

11. Pre-Submission Zero-Hallucination Verification Audit

Prior to final submission, src/verify_submission.py performs comprehensive hard-failure assertions: verifying non-null source URLs, validating that papers without GitHub repos strictly have github_url=null, confirming that star counts are verified integers, and enforcing pricing model and job role enum bounds.

Final Architecture Validation Summary:
- **100% Automated Test Suite Passing:** Strict zero-hallucination, schema conformance, and deterministic entity resolution verified.
- **Full Multi-Tenant Data Isolation:** Complete per-user filesystem segregation and database Row-Level Security enforcement.
- **Synchronized Remote Replication:** Real-time Supabase PostgreSQL persistence and background Google Sheets export.